

LAB MANUAL

GitHub Repository, Kanban Board, Product Backlog and Mini-Sprint

Sample Project: Student Management System

This manual guides students through repository creation, GitHub Issues, a Kanban board, backlog prioritization, story-point estimation, a 3-day mini-sprint, daily updates, and sprint review.

Experiment Focus	Agile project tracking using GitHub Projects
Primary Tools	GitHub Repository, Issues, Projects
Sprint Duration	3-day mini-sprint
Estimated Sprint Scope	4 items, 13 Story Points

Student Name: _____

USN / Roll No.: _____

Date: _____

1. Aim

To initialize a GitHub repository, create and manage tasks using GitHub Issues and a Kanban board, define a Product Backlog with priorities and story-point estimates, and perform a mini-sprint with daily progress updates.

2. Objectives

- Create and initialize a GitHub repository.
- Create project tasks using GitHub Issues.
- Create a Kanban board using GitHub Projects.
- Organize work using Todo, In Progress, and Done.
- Define and prioritize a Product Backlog.
- Estimate work using Story Points.
- Select a Sprint Backlog and conduct a mini-sprint.
- Record daily progress updates and review sprint completion.

3. Software Requirements

- Computer or laptop with internet access.
- Web browser.
- GitHub account.
- Git installed locally - optional for this experiment.

4. Problem Statement

Create a sample **Student Management System** project using GitHub. Initialize a repository, create project tasks using Issues, and organize them using a Kanban board. Define a Product Backlog by assigning priorities and story-point estimates. Select suitable backlog items for a mini-sprint, track their progress through Todo, In Progress, and Done states, record daily updates, and review the sprint at completion.

Workflow Overview

```
Create Repository -> Create Issues -> Create Project Board -> Add Issues  
-> Define Product Backlog -> Set Priority -> Estimate Story Points  
-> Select Sprint Backlog -> Daily Updates -> Sprint Review
```

Part A - Create the GitHub Repository

Step 1: Sign in to GitHub

Open GitHub in a web browser and sign in using your GitHub account.

Step 2: Create a new repository

- Click **+** in the upper-right corner and choose **New repository**.
- Repository name: **student-task-manager**.
- Description: **A sample Student Management System project for demonstrating Agile, Scrum and Kanban practices.**
- Choose **Public**.

- Select **Add a README file**.
- Click **Create repository**.

Expected output: A new repository named **student-task-manager** is created.

Part B - Create GitHub Issues

Go to **Repository -> Issues -> New issue**. Create the following six issues.

Issue 1 - Design Login Page

Title: Design Login Page

Objective: Design a simple login page for the Student Management System.
 Tasks:
 - Create username field.
 - Create password field.
 - Add Login button.
 - Add basic input validation.
 Expected Result: A simple and user-friendly login page should be created.

Issue 2 - Create Student Registration Form

Title: Create Student Registration Form

Objective: Create a form for entering student details.
 Tasks:
 - Add Student Name.
 - Add USN/Roll Number.
 - Add Email.
 - Add Department.
 - Add Semester.
 - Add Submit button.
 - Validate required fields.
 Expected Result: The user should be able to enter student information using the form.

Issue 3 - Create Student Database

Title: Create Student Database

Objective: Create a database for storing student information.
 Tasks:
 - Create Student table.
 - Add Student ID.
 - Add Name.
 - Add USN.
 - Add Email.
 - Add Department.
 - Add Semester.
 - Define suitable data types.
 Expected Result: Student information should be stored correctly in the database.

Issue 4 - Implement Add Student Feature

Title: Implement Add Student Feature

Objective: Implement functionality to add a new student.
 Tasks:
 - Read information from the registration form.
 - Validate student details.
 - Store student information in the database.
 - Display a success message.
 Expected Result: A new student should be successfully added to the system.

Issue 5 - Implement Search Student Feature

Title: Implement Search Student Feature

Objective: Implement functionality to search for an existing student.
 Tasks:
 - Create a search field.
 - Search using USN/Roll Number.
 - Retrieve student information from the database.
 - Display student information.
 - Display a message when the student is not found.
 Expected Result: The correct student information should be displayed.

Issue 6 - Test Student Management System

Title: Test Student Management System

Objective: Test the major functionalities of the Student Management System.
 Tasks:
 - Test Login Page.
 - Test Registration Form.
 - Test Add Student feature.
 - Test Search Student feature.
 - Check input validation.
 - Identify errors or bugs.
 Expected Result: All major functions should operate correctly.

Expected output: The repository contains six GitHub Issues.

Part C - Create the Kanban Board

Step 1: Create a GitHub Project

- Click your profile picture -> **Projects** -> **New project**.
- Choose **Board**.
- Project name: **Student Management System**.
- Click **Create project**.

Step 2: Configure the board

Column	Meaning
Todo	Work has not started.
In Progress	Work is currently being performed.
Done	Work has been completed.

Step 3: Add all six issues to the Todo column

Under **Todo**, click **+ Add item**, search for each issue, and add it to the project.

Todo	In Progress	Done
Design Login Page		
Create Student Registration Form		
Create Student Database		
Implement Add Student Feature		
Implement Search Student Feature		
Test Student Management System		

Checkpoint: All six tasks should initially be in Todo.

Part D - Define the Product Backlog

A **Product Backlog** is the complete prioritized list of work required for the project.

ID	Product Backlog Item	Priority	Story Points
PB1	Design Login Page	High	2
PB2	Create Student Registration Form	High	3
PB3	Create Student Database	High	5
PB4	Implement Add Student Feature	High	3
PB5	Implement Search Student Feature	Medium	3
PB6	Test Student Management System	High	5

Part E - Add the Priority Field

Create a table view in the project: **+ New view -> Table**.

Add a new field named **Priority** with field type **Single select**. Create the options **High**, **Medium**, and **Low**.

```
Design Login Page -> High
Registration Form -> High
Create Database -> High
Add Student -> High
Search Student -> Medium
Testing -> High
```

Part F - Add Story Point Estimation

Create another field named **Story Points** with field type **Number**.

Task	Story Points
Design Login Page	2
Registration Form	3

Task	Story Points
Create Database	5
Add Student	3
Search Student	3
Testing	5

Story Points estimate relative effort and complexity. They are not hours.

Story Points	Interpretation
1	Very Easy
2	Easy
3	Medium
5	Difficult
8	Very Difficult

Part G - Define Sprint 1

For this experiment, perform a **3-day mini-sprint**.

Sprint Goal: Develop the basic student registration functionality of the Student Management System.

Sprint Backlog

Sprint 1 Task	Story Points
Design Login Page	2
Create Student Registration Form	3
Create Student Database	5
Implement Add Student Feature	3
Total	13

The remaining two items, **Implement Search Student Feature** and **Test Student Management System**, stay in the Product Backlog for a future sprint.

Part H - Perform the Mini-Sprint

Day 1

Move **Design Login Page** from **Todo** -> **In Progress**.

Daily Update - Day 1

Completed:

- Sprint planning and task estimation.

Working on:

- Designing the Login Page.
- Creating username and password fields.
- Adding the Login button.
- Performing basic input validation.

Blockers:

- None.

After completion, move **Design Login Page** from **In Progress** -> **Done**.

Todo	In Progress	Done
Registration Form		Design Login Page
Database		
Add Student		
Search Student		
Testing		

Day 2

Move **Create Student Registration Form** and **Create Student Database** from **Todo** -> **In Progress**.

Registration Form daily update

Daily Update - Day 2

Completed:

- Design Login Page.

Working on:

- Creating the Student Registration Form.
- Adding Name, USN, Email, Department and Semester fields.
- Adding the Submit button.
- Validating required fields.

Blockers:

- None.

Database daily update

Daily Update - Day 2

Completed:

- Design Login Page.

Working on:

- Creating the Student Database.
- Creating the Student table.
- Adding Student ID, Name, USN, Email, Department and Semester fields.
- Defining suitable data types.

Blockers:

- None.

After completion, move both tasks to **Done**.

Todo	In Progress	Done
Add Student		Login Page
Search Student		Registration Form
Testing		Database

Day 3

Move **Implement Add Student Feature** from **Todo** -> **In Progress**.

Daily Update - Day 3

Completed:

- Design Login Page.
- Student Registration Form.
- Student Database.

Working on:

- Implementing the Add Student feature.
- Connecting the registration form to the database.
- Validating student information.
- Storing student details in the database.

Blockers:

- None.

After completion, move **Implement Add Student Feature** to **Done**.

Part I - Final Kanban Board

Todo	In Progress	Done
Implement Search Student Feature		Design Login Page
Test Student Management System		Create Student Registration Form
		Create Student Database
		Implement Add Student Feature

Metric	Value
Todo	2
In Progress	0

Metric	Value
Done	4

Part J - Sprint Review

Sprint Review Item	Value
Sprint Goal	Develop the basic student registration functionality.
Planned Work	4 tasks
Planned Story Points	13 SP
Completed Tasks	4
Completed Story Points	13 SP
Blockers	None
Sprint Result	Sprint Goal Achieved

Part K - GitHub Project Status Update

Click **Add status update** and select **Complete**.

Sprint 1 Completed

Sprint Goal:
Develop the basic student registration functionality.

Completed Tasks:
- Design Login Page
- Create Student Registration Form
- Create Student Database
- Implement Add Student Feature

Planned Story Points: 13
Completed Story Points: 13

Blockers:
None

Result:
All tasks selected for Sprint 1 were completed successfully.

Part L - Screenshots to Include in the Lab Record

- GitHub repository creation.
- Issues page showing all six issues.
- Initial Kanban board with all tasks in Todo.
- Product Backlog table showing Priority and Story Points.
- At least one Day 1, Day 2, and Day 3 daily update comment.
- Final Kanban board showing 2 Todo, 0 In Progress, and 4 Done.
- Sprint project status marked Complete.

Observation Table

Activity	Observation
Repository created	Successful
Issues created	6
Kanban columns	Todo, In Progress, Done
Product Backlog items	6
Sprint 1 selected items	4
Planned Story Points	13
Completed Story Points	13
Remaining backlog items	2
Sprint status	Complete

Result

A GitHub repository was successfully initialized for the Student Management System project. GitHub Issues were used to define project tasks and a Kanban board was created using Todo, In Progress, and Done states. A Product Backlog was defined by assigning priorities and story-point estimates. Four backlog items totaling 13 story points were selected for a three-day mini-sprint. Daily updates were recorded and the tasks were moved through the Kanban workflow. All 13 story points planned for Sprint 1 were successfully completed.

Viva Questions

1. What is Git?
2. What is GitHub?
3. What is a GitHub repository?
4. What is a GitHub Issue?
5. What is a Kanban board?
6. What are the three basic columns used in this experiment?
7. What is a Product Backlog?

8. What is a Sprint Backlog?
9. What is a Sprint?
10. What are Story Points?
11. Are Story Points equal to hours?
12. What is the purpose of task priority?
13. What is the difference between Todo and In Progress?
14. What is a daily update?
15. What is a blocker?
16. What is Sprint Planning?
17. What is a Sprint Review?
18. Why are all Product Backlog items not necessarily selected for a sprint?
19. How many Story Points were selected for Sprint 1?
20. What happened to the tasks that were not selected for Sprint 1?

Quick Memory Line

Repository stores the project -> Issues define tasks -> Product Backlog organizes all work -> Story Points estimate effort -> Sprint selects work -> Kanban tracks progress -> Daily updates record progress -> Sprint Review evaluates completion.